

# Chapter 9

1. What index value does the third element of an array have?

The value of the third element in an array is 2 [ 0, 1, 2, 3]

2. Write the declaration for an array named quantities that stores 20 integers

```
int[] quantities = new int[20];
```

3. Write a declaration for an array named heights storing the numbers 1.65, 2.15, and 4.95

```
double[] heights = {1.65, 2.15, 4.95}
```

4. Write a for-each statement that displays the integer values stored in an array named grades

```
for (int grade : grades) {  
    System.out.println(grade);
```

6. How does passing an entire array to a method differ from passing a single element of the array?

When you pass an entire array, the method can access the whole array. But when you pass a single element in the array, the method can only access that single element.

7. Why are offset array indexes required in some cases?

Offset array indexes are required in some cases when you want to use a number from a specific position in the array instead of starting from the beginning of the array, such as skipping the first couple of numbers in an array

8. What output is displayed by the statements below?

```
String name = "Elaine";  
System.out.println(name.charAt(3));
```

"i" because it is the 3rd character

